COST AND PERFORMANCE · DOMAIN 6 - COST & PERFORMANCE OPTIMIZATION

Managed tables and predictive optimization

Section 6. Why the maintenance job you were taught to write is now the wrong answer.

SECTION 6 of 10 WEIGHT 13% TOPIC Managed tables

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The question is not where the files live. It is **who is responsible for keeping them tidy**.

Managed, external and foreign

	WHO MANAGES WHAT
Managed	Unity Catalog manages lifecycle, location and optimisations. Automatic optimisations: yes.
External	You specify the location and manage the lifecycle. Automatic optimisations: limited.
Foreign	An external system manages data and metadata. Automatic optimisations: none.

ON THE EXAM

Managed is described in the documentation as the default and recommended table type. That phrasing matters when a question asks what you should use.

Five things, and the last one is underrated

- **Automatic optimisation**, reducing both storage and compute costs.
- **Faster queries across all client types**, not only from Databricks.
- **Automatic table maintenance**, which is the maintenance job you no longer write.
- Secure access for non-Databricks clients through open APIs.
- **Automatic upgrades to the latest platform features.**

ON THE EXAM

That last one compounds. An external table stays on the feature set it was created with unless someone actively migrates it.

What happens when you drop it

- Dropping a **managed** table deletes the metadata **and the underlying data files**.
- Dropping an **external** table removes only the catalog metadata. The files remain.
- That is the safety argument people make for external tables, and it is a real one.
- It is also how organisations accumulate orphaned data nobody is paying attention to but everybody is paying for.

> ON THE EXAM
>
> Both halves are true. External protects you from an accidental drop, and leaves you paying for data that stopped being useful months ago.

Governed, but not optimised

- Foreign tables are registered through **query federation or catalog federation**, from module 16.
- Foreign tables backed by Delta Lake **lack many optimisations** available to managed tables.
- Databricks recommends migrating to managed tables for **production workloads or frequently queried datasets**.
- So federation is for reaching data. It is not a performance strategy.

> ON THE EXAM
>
> This is module 16's recommendation from the other direction. Federation gives you governance and access, not speed.

Predictive optimization runs **OPTIMIZE, VACUUM and ANALYZE** for you, deciding which tables need them and when.

The rollout, in dates

- **Enabled by default** for accounts created on or after **11 November 2024**.
- Existing accounts are being enabled by gradual rollout, **expected to complete by August 2026**.
- It runs **only on Unity Catalog managed tables**. An external table gets nothing from it.
- It is **not available in all regions**.

> ON THE EXAM
>
> The dates matter for a professional exam because they explain why two teams can describe completely different behaviour on the same platform.

Predictive optimization does not run Z-order

- **OPTIMIZE does not run ZORDER when executed by predictive optimization.**
- On a table that uses Z-order, predictive optimization **ignores the Z-ordered files**.
- So a Z-ordered table receives **less** automatic benefit than an unclustered one, not more.
- That is one of the clearest practical arguments for moving from Z-order to liquid clustering.

> ON THE EXAM
>
> Counterintuitive and precisely stated. A table you carefully Z-ordered years ago is now partly excluded from automatic maintenance.

Set retention before you enable it

- Vacuum retention comes from the **deleted file retention duration** table property, defaulting to **seven days**.
- Vacuum removes data files no longer referenced within that window, which **limits how far back time travel reaches**.
- To keep longer history, **set the property before enabling** predictive optimization.
- That is a deliberate decision, not a default to accept quietly, if audit or reproducibility matters.

> ON THE EXAM
>
> Automatic maintenance and long time travel are in tension. The property is where you resolve it, and the docs say to set it first.

How predictive optimization is billed

- It runs on **serverless compute for jobs**, and your account is billed for that compute.
- It appears under the **serverless jobs SKU**, shared with other products.
- So it shows up as a cost line that no job in your schedule explains.
- Break it out with the **billing origin product** column, exactly as module 17 described.

> ON THE EXAM
>
> Module 17's shared-SKU trap and this module's automatic maintenance are the same story from two directions.

Automatic liquid clustering

- Set **CLUSTER BY AUTO** on a managed unpartitioned or liquid table.
- Predictive optimization then **analyses how the table is queried** and selects clustering keys itself.
- It **evaluates whether the performance gain outweighs the clustering overhead** before changing anything.
- Keys can be updated over time as query patterns change, which manual clustering never does.

> ON THE EXAM
>
> This is the argument that layout decisions age badly. A key chosen correctly two years ago may be wrong now, and nobody revisits it.

Finding out what it decided

- **DESCRIBE EXTENDED** shows the reasons predictive optimization **skipped** a table.
- Typical reasons: the table is **already well-clustered**, or **too small to benefit** from compaction.
- Operation history is also tracked in a **system table**, with 180-day retention from module 17.
- So 'it did nothing' is usually a decision rather than a failure, and you can read the reason.

> ON THE EXAM
>
> When a question describes automatic maintenance apparently not running, the first move is to check why it chose not to.

Converting an external table to managed

- Use **ALTER TABLE SET MANAGED**. Requires DBR 17.0 or above, or serverless.
- Preferred over a create-table-as-select because it **retains history**, keeps permissions and views, and handles concurrent writes.
- It also supports **rolling back** to external, and minimises reader and writer downtime.
- **Cancel any running OPTIMIZE jobs first**, and schedule none during the conversion.

> ON THE EXAM
>
> Retaining history is the deciding advantage. A create-table-as-select gives you a new table with no past, and no time travel.

Two behaviours worth knowing

- From **DBR 18.1**, path-based reads and writes to the old external location **automatically redirect** to the managed location.
- That lets legacy code using storage paths keep working, at a slight performance cost.
- **Foreign table conversions do not redirect**, and predictive optimization is set to **inherit** rather than enabled.
- After a foreign conversion, **restart any streaming jobs** reading or writing the table.

> ON THE EXAM
>
> Databricks still recommends moving to name-based access. Redirect is a migration aid, not a destination.

What to carry into the exam

- **Managed is the default and recommended type.** External gives you the location and the maintenance.
- Dropping managed deletes the files. Dropping external leaves them, and the bill.
- Predictive optimization runs **OPTIMIZE, VACUUM and ANALYZE** on managed tables only.
- It **does not run ZORDER**, and **ignores Z-ordered files**. Z-order gets less automatic benefit, not more.
- Convert with **SET MANAGED** to retain history. Set vacuum retention **before** enabling.

> ON THE EXAM
>
> When an option proposes a scheduled weekly OPTIMIZE and VACUUM job on a managed table, ask what predictive optimization was already doing.

> A team maintains a weekly job that runs OPTIMIZE and VACUUM across forty Unity Catalog managed tables. Costs for that job have been questioned, and a review notes that query performance has not changed since predictive optimization was enabled on the account.
>
> The team argues the job is essential and should be kept.

What is the best assessment?

A. Keep the job. Predictive optimization supplements manual maintenance but does not replace it.

B. The job duplicates work predictive optimization already performs on managed tables. Retire it and verify with DESCRIBE EXTENDED and the system table.

C. Keep the job but reduce it to monthly, to halve the cost.

D. Convert the tables to external so the manual job remains meaningful.

◆ PAUSE AND DECIDE

What is the best assessment?

A Keep the job. Predictive optimization supplements manual maintenance but does not replace it.

It removes the need to manually manage maintenance operations on managed tables. That is its stated purpose.

B The job duplicates work predictive optimization already performs on managed tables. Retire it and verify with DESCRIBE EXTENDED and the system table.

All forty tables are managed, so the operations are already being run when they are worth running.

C Keep the job but reduce it to monthly, to halve the cost.

Halves the cost of work that is entirely redundant.

D Convert the tables to external so the manual job remains meaningful.

Gives up automatic maintenance to justify a job. The reasoning runs backwards.

ON THE EXAM

The evidence is in the scenario: performance did not change when predictive optimization arrived, because the work was already being done.

> Two teams run identical workloads on identical data. Team A reports that maintenance happens automatically and they have written no maintenance jobs. Team B reports that nothing happens automatically and they schedule OPTIMIZE weekly.
>
> Both use Unity Catalog managed tables. Team B's account was created in 2021; Team A's in 2025.

What is the most likely explanation?

A. Team B's tables are external despite what they believe.

B. Predictive optimization is default-on for accounts created from November 2024, and older accounts are still being enabled by a gradual rollout.

C. Team B has fewer tables, so predictive optimization judges them not worth optimising.

D. Predictive optimization requires an explicit table property that Team B has not set.

◆ PAUSE AND DECIDE

What is the most likely explanation?

A Team B's tables are external despite what they believe.

Possible in general, and the account dates are the detail the scenario actually gives you.

B Predictive optimization is default-on for accounts created from November 2024, and older accounts are still being enabled by a gradual rollout.

The account creation dates straddle the cutover exactly. This is why two teams describe different platforms.

C Team B has fewer tables, so predictive optimization judges them not worth optimising.

The workloads and data are stated to be identical.

D Predictive optimization requires an explicit table property that Team B has not set.

It is an account-level default, not a per-table property to opt into.

ON THE EXAM

The account creation dates are the only asymmetry in the scenario. When a question supplies dates, they are usually the answer.

> An engineer notices that one large managed table receives far less automatic maintenance than similar tables, despite being heavily queried. The table was carefully Z-ordered on two columns three years ago and the layout has not been revisited.
>
> Predictive optimization is enabled and working on other tables in the same schema.

What explains this?

A. Predictive optimization does not run ZORDER, and ignores Z-ordered files on tables that use it.

B. Z-ordered tables are already optimal, so no maintenance is required.

C. The table exceeds a size threshold above which predictive optimization does not operate.

D. Z-order must be re-run manually before predictive optimization will consider the table.

◆ PAUSE AND DECIDE

What explains this?

A Predictive optimization does not run ZORDER, and ignores Z-ordered files on tables that use it.

Stated directly. The Z-ordering is what excludes those files from automatic maintenance.

B Z-ordered tables are already optimal, so no maintenance is required.

A layout chosen three years ago against today's query patterns is unlikely to be optimal.

C The table exceeds a size threshold above which predictive optimization does not operate.

Skipping happens for tables too small to benefit, not too large.

D Z-order must be re-run manually before predictive optimization will consider the table.

Re-running Z-order produces more Z-ordered files to ignore.

ON THE EXAM

The engineering effort is causing the problem. Moving to liquid clustering, or automatic liquid clustering, restores full automatic maintenance.

> A regulated team must be able to reproduce the exact contents of a table as at any date in the past 90 days. They are about to migrate the table to Unity Catalog managed and enable predictive optimization.
>
> Nobody has changed any table properties.

What must they do?

A Nothing. Managed tables retain full history indefinitely.

B Set the deleted file retention duration to cover 90 days before enabling predictive optimization.

C Disable VACUUM entirely on the table after migration.

D Keep the table external, since managed tables cannot support extended time travel.

◆ PAUSE AND DECIDE

What must they do?

A Nothing. Managed tables retain full history indefinitely.

Retention defaults to seven days. Files outside it are removed by VACUUM.

B Set the deleted file retention duration to cover 90 days before enabling predictive optimization.

The documented sequence: set the property first, so automatic VACUUM respects the longer window.

C Disable VACUUM entirely on the table after migration.

Trades one problem for unbounded storage growth, and the property already expresses the requirement.

D Keep the table external, since managed tables cannot support extended time travel.

They support it. It is a property setting, not a table type limitation.

ON THE EXAM

Note the ordering in the answer. The documentation says to set the property before enabling, not afterwards.

> A team must convert 200 external Delta tables to managed. Requirements: retain time travel history, keep existing permissions and views, and avoid breaking legacy notebooks that read tables by storage path.
>
> An engineer proposes a create-table-as-select into new managed tables, then dropping the old ones.

What is the better approach, and why?

A The CTAS approach is fine; it produces clean tables with no legacy baggage.

B Use ALTER TABLE SET MANAGED, which retains history and permissions and redirects path-based access on recent runtimes.

C Use DEEP CLONE into a managed table, which copies history.

D Leave them external and enable predictive optimization on them individually.

◆ PAUSE AND DECIDE

What is the better approach, and why?

A The CTAS approach is fine; it produces clean tables with no legacy baggage.

It also produces tables with no history, and permissions and views must be rebuilt by hand.

B Use ALTER TABLE SET MANAGED, which retains history and permissions and redirects path-based access on recent runtimes.

Meets all three requirements, and is the documented recommendation over CTAS.

C Use DEEP CLONE into a managed table, which copies history.

Deep clone copies data and creates a new table. Permissions, views and path-based access still break.

D Leave them external and enable predictive optimization on them individually.

Predictive optimization runs only on managed tables. There is nothing to enable.

ON THE EXAM

Three requirements, and only one option meets all three. The path-based redirect is the one people do not know exists.

> A cost review flags a growing spend line that no scheduled job accounts for. Investigation shows it under the serverless jobs SKU. The team runs no serverless jobs at all.
>
> They recently migrated 300 tables to Unity Catalog managed.

What is the most likely explanation?

A An unauthorised serverless job is running in the account.

B Predictive optimization runs on serverless compute for jobs and bills under that SKU. Break it out by billing origin product.

C Managed tables are billed for storage under the serverless jobs SKU.

D The migration itself is still running and will stop billing when complete.

◆ PAUSE AND DECIDE

What is the most likely explanation?

A An unauthorised serverless job is running in the account.

A reasonable worry, and the documented explanation fits the facts far better.

B Predictive optimization runs on serverless compute for jobs and bills under that SKU. Break it out by billing origin product.

The migration made 300 tables eligible, and maintenance now runs automatically on all of them.

C Managed tables are billed for storage under the serverless jobs SKU.

Storage and compute are billed separately.

D The migration itself is still running and will stop billing when complete.

Growing spend suits ongoing maintenance rather than a finite migration.

ON THE EXAM

This is module 17's shared-SKU scenario with its cause supplied. The spend is real, expected, and usually cheaper than what it replaced.